

THUMULA VAMSHIDHAR RAO

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SUMMARY

Full-stack engineer who builds agentic AI systems and real-time applications end to end – backend, orchestration, and the prompts that hold up in production. Comfortable across LLM orchestration frameworks (LangGraph, Agno), retrieval and knowledge-graph systems, real-time communication (Socket.IO, WebSockets), and standard backend infra (Node.js, FastAPI, PostgreSQL, Docker). I like owning things end to end rather than a single layer of the stack – from wiring the orchestration logic and prompts, through the backend and data layer, to the operational edges (auth, rate limiting, file handling) that most people leave for later. Currently looking for a role where I can take real, unclear problems and ship something that works against a live system, not just a demo.

TECHNICAL SKILLS

Agentic AI & Orchestration: LangGraph, Agentic RAG, knowledge graphs (NetworkX), tool calling, prompt engineering, structured output contracts

Real-Time & Backend: Node.js, Express.js, Socket.IO (WebSockets), Python, FastAPI, REST APIs, JWT auth

Data & Infra: PostgreSQL, Prisma, SQLAlchemy, AWS S3, Docker, Docker Compose, Git

PROJECTS

Marketing AI Agent | *Python, LangGraph, Gemini, Chroma, NetworkX* 2025

- Built a multi-platform marketing copilot on a LangGraph workflow that classifies each query and routes it through retrieval, knowledge-graph lookup, or copy generation – Q&A vs. platform-optimized ad copy for Facebook, Instagram, Google Ads, LinkedIn, Twitter, and TikTok.
- Combined agentic RAG (Chroma + BGE embeddings) with a hand-built knowledge graph encoding per-platform constraints (character limits, tone fit, CTA mapping) so generated copy respects real platform rules, not just LLM judgment.
- Wrote and iterated prompts against a measurable rubric – platform compliance, tone match, CTA presence, and a hallucination check comparing numeric claims to sources – with a feedback loop that promotes highly-rated copy into reusable patterns.

NexChat – Real-Time Communication Platform | *Node.js, TypeScript, Socket.IO, PostgreSQL* | ([GitHub](#)) 2025

- Built a full-stack real-time chat app: WebSocket messaging via Socket.IO with JWT-authenticated connections, room-based broadcasting, and persistent message history on PostgreSQL/Prisma.
- Handled the operational edges directly – rate limiting, input validation, and secure file sharing via pre-signed S3 URLs – rather than leaving them as follow-up work.

OpenEnv – Microservice Incident-Debugging Simulator for AI Agents | *Python, FastAPI, Docker* | ([GitHub](#)) 2026

- Built an HTTP environment server that lets AI agents take multi-step, tool-driven actions against a simulated system with realistic service dependencies, then grades their diagnosis and recovery after injected partial failures.
- Finalist, Meta PyTorch OpenEnv Hackathon × Scaler School of Technology.

EXPERIENCE

AI/ML Intern – Ebani Tech Pvt. Ltd. June 2024

Hyderabad (Onsite)

- Built a GenAI-powered conversational assistant letting clients query ad performance metrics in natural language, and integrated ML-derived insights into backend APIs consumed by business teams.

ACHIEVEMENTS

Meta PyTorch OpenEnv Hackathon × Scaler School of Technology – Finalist 2026

- Reached the final round with **OpenEnv**, an AI agent evaluation platform for benchmarking reasoning and recovery in simulated environments, selected out of a broad hackathon field.

EDUCATION

Vellore Institute of Technology

Bachelor of Technology — GPA: 8.73 / 10.00

Vellore, Tamil Nadu

Sep 2022 – May 2026